Practical 11: Student Management CRUD with Search and Filter
Problem Definition

Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.

Key question
1) Are SELECT, INSERT, UPDATE, and DELETE operations working?yes, SELECT, INSERT, UPDATE, and DELETE operations are implemented using MySQLi.

2. Is search by name/email/course implemented?
Partially. Search by name and email is implemented, and filtering by course is implemented.

3. Are delete operations confirmed before execution?
Partially. Search by name and email is implemented, and filtering by course is implemented.

Code:

Db.php

```php
<?php
$conn = new mysqli("localhost", "root", "", "student_management");

if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}

$conn->set_charset("utf8mb4");
?>
```

Edit.php

```php
<?php
include 'db.php';

$id = (int)($_GET['id'] ?? $_POST['id'] ?? 0);
$msg = "";
```

```
$stmt = $conn->prepare("SELECT * FROM students WHERE id=?");
$stmt->bind_param("i", $id);
$stmt->execute();
$student = $stmt->get_result()->fetch_assoc();

if (!$student) die("Student not found!");

if (isset($_POST['update'])) {
    $name = trim($_POST['name']);
    $roll = trim($_POST['roll_no']);
    $email = trim($_POST['email']);
    $course = $_POST['course'];

    if ($name && $roll && filter_var($email, FILTER_VALIDATE_EMAIL)
        && in_array($course, ['CSE','IT','CE'])) {

        $stmt = $conn->prepare(
            "UPDATE students SET name=?,roll_no=?,email=?,course=? WHERE
id=?"
        );
        $stmt->bind_param("ssssi", $name, $roll, $email, $course, $id);

        if ($stmt->execute()) {
            header("Location: index.php");
            exit;
        } else {
            $msg = "Update Failed!";
        }
    } else {
        $msg = "Please enter valid details!";
    }
}
?>

<h2>Edit Student</h2>
<p><?php echo htmlspecialchars($msg); ?></p>

<form method="post">
    <input type="hidden" name="id" value="<?php echo $id; ?>">
```

```
    <input name="name" required
      value="<?php echo htmlspecialchars($_POST['name'] ??
$student['name']); ?>">

    <input name="roll_no" required
      value="<?php echo htmlspecialchars($_POST['roll_no'] ??
$student['roll_no']); ?>">

    <input type="email" name="email" required
      value="<?php echo htmlspecialchars($_POST['email'] ??
$student['email']); ?>">

    <select name="course" required>
        <?php foreach (['CSE','IT','CE'] as $c) { ?>
        <option value="<?php echo $c; ?>"
          <?php echo (($_POST['course'] ?? $student['course']) === $c) ?
'selected' : ''; ?>>
          <?php echo $c; ?>
        </option>
        <?php } ?>
    </select>

    <button name="update">Update Student</button>
</form>

<a href="index.php">Back</a>
```

Index.php

```
<?php
include 'db.php';

$id = (int)($_GET['id'] ?? $_POST['id'] ?? 0);
$msg = "";

$stmt = $conn->prepare("SELECT * FROM students WHERE id=?");
$stmt->bind_param("i", $id);
$stmt->execute();
```

```php
$student = $stmt->get_result()->fetch_assoc();

if (!$student) die("Student not found!");

if (isset($_POST['update'])) {
    $name = trim($_POST['name']);
    $roll = trim($_POST['roll_no']);
    $email = trim($_POST['email']);
    $course = $_POST['course'];

    if ($name && $roll && filter_var($email, FILTER_VALIDATE_EMAIL)
        && in_array($course, ['CSE','IT','CE'])) {

        $stmt = $conn->prepare(
            "UPDATE students SET name=?,roll_no=?,email=?,course=? WHERE
id=?"
        );
        $stmt->bind_param("ssssi", $name, $roll, $email, $course, $id);

        if ($stmt->execute()) {
            header("Location: index.php");
            exit;
        } else {
            $msg = "Update Failed!";
        }
    } else {
        $msg = "Please enter valid details!";
    }
}
?>

<h2>Edit Student</h2>
<p><?php echo htmlspecialchars($msg); ?></p>

<form method="post">
    <input type="hidden" name="id" value="<?php echo $id; ?>">

    <input name="name" required
      value="<?php echo htmlspecialchars($_POST['name'] ??
$student['name']); ?>">
```

```
    <input name="roll_no" required
      value="<?php echo htmlspecialchars($_POST['roll_no'] ??
$student['roll_no']); ?>">

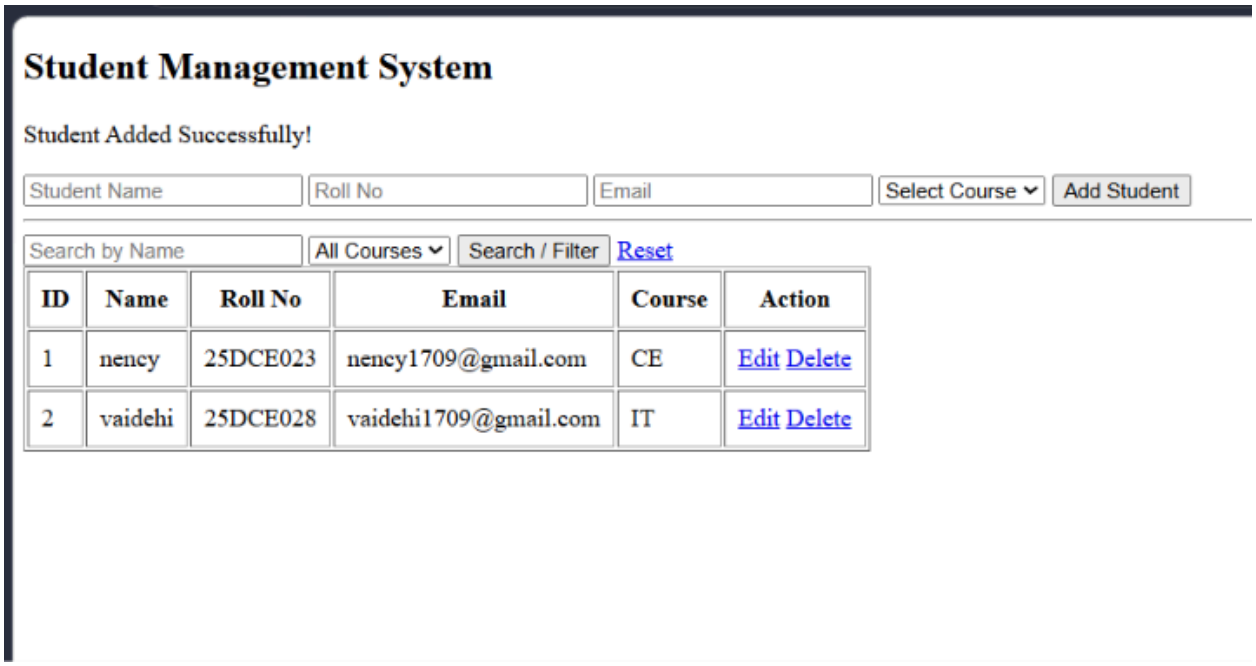
    <input type="email" name="email" required
      value="<?php echo htmlspecialchars($_POST['email'] ??
$student['email']); ?>">

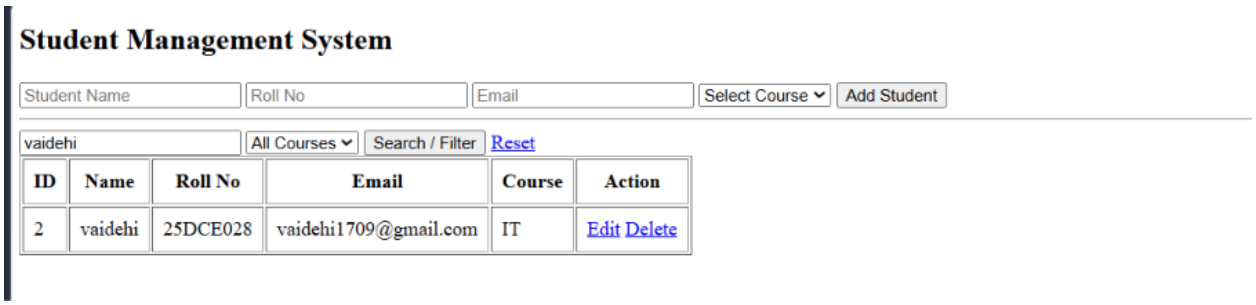
    <select name="course" required>
        <?php foreach (['CSE','IT','CE'] as $c) { ?>
        <option value="<?php echo $c; ?>"
          <?php echo (($_POST['course'] ?? $student['course']) === $c) ?
'selected' : ''; ?>>
          <?php echo $c; ?>
        </option>
        <?php } ?>
    </select>

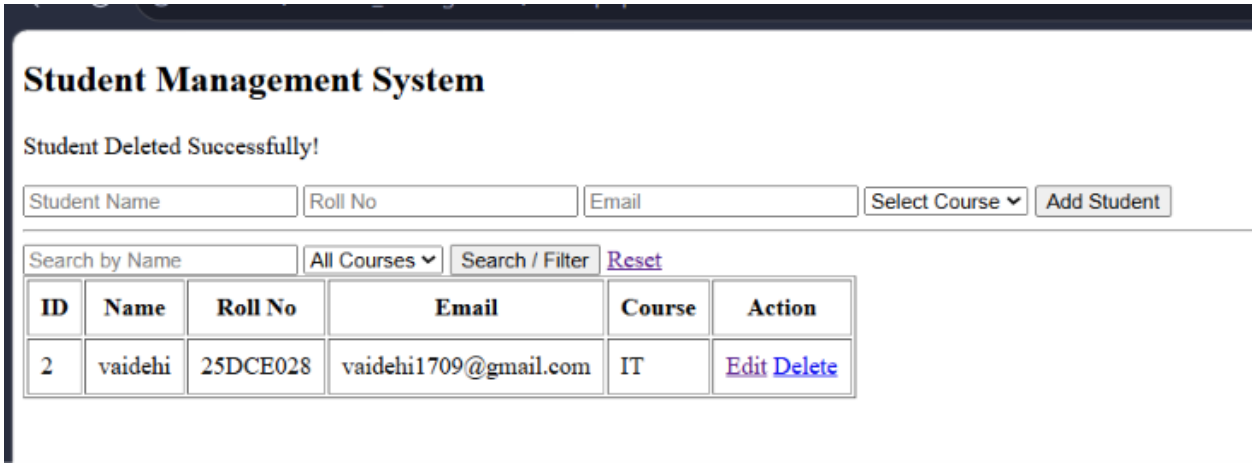
    <button name="update">Update Student</button>
</form>

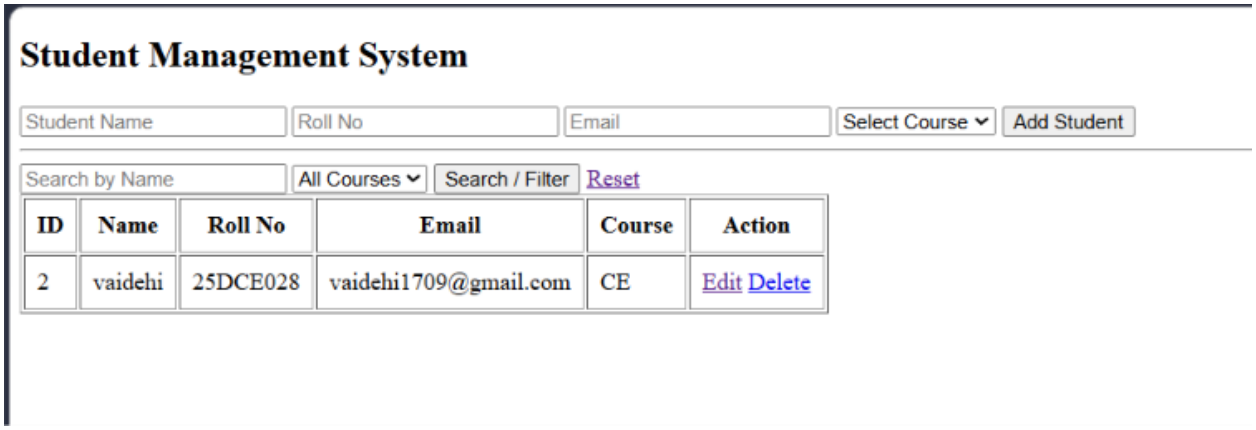
<a href="index.php">Back</a>
```

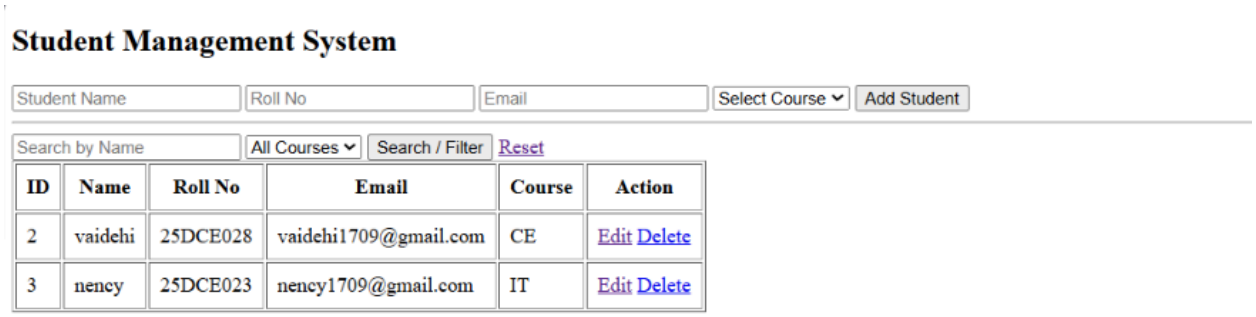
Output


Student Management System
Student Added Successfully!
Student Name
Roll No
Email
Select Course
Add Student
Search by Name
All Courses
Search / Filter
Reset
ID Name Roll No Email Course Action
1 nency 25DCE023 nency1709@gmail.com CE Edit Delete
2 vaidehi 25DCE028 vaidehi1709@gmail.com IT Edit Delete


Student Management System
Student Name
Roll No
Email
Select Course
Add Student
vaidehi
All Courses
Search / Filter
Reset
ID Name Roll No Email Course Action
2 vaidehi 25DCE028 vaidehi1709@gmail.com IT Edit Delete


Student Management System
Student Deleted Successfully!
Student Name
Roll No
Email
Select Course
Add Student
Search by Name
All Courses
Search / Filter
Reset
ID Name Roll No Email Course Action
2 vaidehi 25DCE028 vaidehi1709@gmail.com IT Edit Delete


Student Management System
Student Name
Roll No
Email
Select Course
Add Student
Search by Name
All Courses
Search / Filter
Reset
ID
Name
Roll No
Email
Course
Action
2
vaidehi
25DCE028
vaidehi1709@gmail.com
CE
Edit
Delete


Student Management System
Student Name
Roll No
Email
Select Course
Add Student
Search by Name
All Courses
Search / Filter
Reset
ID
Name
Roll No
Email
Course
Action
2
vaidehi
25DCE028
vaidehi1709@gmail.com
CE
Edit
Delete
3
nency
25DCE023
nency1709@gmail.com
IT
Edit
Delete